Richard Edmond Victor Sesay

Environmental Engineer & Materials Sustainability Researcher

Tartu, Estonia | rev.sesay_341@proton.me | ORCID 0009-0004-0700-4386 | ResearchGate: Richard-Sesay

PROFILE

Environmental engineer (MEng, Tongji University) with a First Division BSc in Biology and Chemistry and peer-reviewed publications on solid waste management, circular economy, microplastics and water and environmental quality. Currently an **EIT AMIR** Master's candidate (University of Tartu, then University of Bordeaux, 2026–2028), specialising in critical raw materials recovery and circular economy. Hands-on experience in analytical chemistry, environmental sampling, GIS mapping and multivariate data modelling in Sierra Leone, China and Estonia, plus science communication for a KIK-funded education project. Seeking part-time or research-adjacent roles in Estonia in environmental science, sustainability, materials recycling or laboratory and field research support.

EDUCATION

MSc, Advanced Materials – Innovative Recycling (EIT AMIR) – Tartu / Bordeaux 2026–2028 (expected)
EIT-labelled programme. EIT AVSA (Added Value Student Activity) Grant, EUR 15,000.

MEng, Environmental Science – Tongji University, Shanghai 2023–2025
Distinction, 89.24/100. Fully funded MOFCOM Chinese Government Scholarship.

BSc.Ed, Biology and Chemistry – Milton Margai Technical University, Sierra Leone 2017–2021
First Division (First Class equivalent), GPA 4.31/5.00.

EXPERIENCE

Researcher & Science Communicator – Eesti Roheline Liikumine (ERL) 2026–present
- Prepared a research dossier with sourced figures and a 10-minute animated video script on the circular economy of disposable e-cigarettes for a KIK-funded youth education project.

HR and Organisational Development Intern – Estonian Student Satellite Foundation, Tartu Apr–Jun 2026
- Supported member onboarding and organisational development; coordinated engineers and researchers within Estonian legal frameworks for IP and NDAs.

Project Assistant – National Protected Area Authority (GEF-IUCN-MRU), Sierra Leone 2020–2023
- Conducted field surveys across 10,000+ hectares of protected forest, analysing data in R and SPSS.
- Produced 10+ GIS maps of biodiversity and land-use change; coordinated 20+ community groups and trained 250+ households.

Biology and Chemistry Teacher – Wamagriso Senior Secondary School, Sierra Leone 2018–2019
- Taught 300+ students and supervised 150+ in laboratory safety; ran environmental outreach for 200+ residents.

SKILLS

Laboratory	qPCR, DNA extraction, FISH, flow cytometry, HPLC, water and wastewater analysis, environmental sampling
Software, analysis	Python, R, STATA, SPSS, OriginPro; multivariate modelling
GIS	QGIS, ArcGIS
Languages	English (native), Chinese (A2)

SELECTED PUBLICATIONS (FULL LIST ON ORCID AND RESEARCHGATE)

- Co-author (2nd of 5). *Biodegradation of pharmaceutical contaminants in wastewater using microbial consortia: Mechanisms, applications, and challenges.* J. Environ. Manage. (2025). doi:10.1016/j.jenvman.2025.125564
- First author. *Circular Economy in Municipal Solid Waste Management: Innovations and Challenges for Urban Sustainability.* J. Environ. Prot. (2025). doi:10.4236/jep.2025.162003
- First author. *Advancing Micro Plastic Analysis: A Comprehensive Review of Detection and Characterization Techniques.* Asian J. Environ. Ecol. 23(12) (2024). doi:10.9734/ajee/2024/v23i12636
- First author. *Invasive Species and Biodiversity: Mechanisms, Impacts, and Strategic Management for Ecological Preservation.* Asian J. Environ. Ecol. 23(9) (2024). doi:10.9734/ajee/2024/v23i9600
- First author. *Invisible Invaders: Unmasking the Emerging Threats of Volatile Organic Compounds in Indoor Air.* Int. J. Latest Eng. Res. Appl. 9(7), 34–41. doi:10.56581/ijlera.9.7.34-41

AWARDS

EIT Added Value Student Activity Grant (EUR 15,000, 2026); MOFCOM Chinese Government Scholarship, Tongji University; First-Division and Top-Excelling Student, MMTU.